



Department of Computer Engineering
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Pacman Game (Java)

Console-Based

Implementation

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Course: CMPE-211L(Object Oriented
Programming)

Overview:

This project is simplified console-based implementation of a Pacman game, made with Java .Game consists of main class that acts as game controller, grid-type game board and several objects that interact with each other..

User controls pacman character through keyboard input, aiming to eat all food objects while avoiding ghost objects (which move randomly). Game ends when either all food objects are eaten (win condition) or the ghost and pacman collide(lose condition)

Project Structure:

The system is designed on object-oriented programming principles. It contains following classes:

Pacman Game(Main class):

- This is the central control panel of game.
- It handles user input.

- It contains all the objects created. It runs the game loop and constantly checks winning and losing conditions.

Gameboard:

- The gameboard builds the grid where the game will function. It displays after every loop with updated positions.
- It controls the grid dimensions; width and height.
- For simplicity the objects to display are represented using single characters “P” for pacman, “G” for ghosts and “*” for food.

Food class:

- It is used to create food objects on grid.
- It initializes and stores food coordinates as random positions on board.
- It tracks whether food is eaten or not, which is later used for win condition.

Ghost class:

- It stores the x and y coordinates of each ghost object.
- The ghost moves randomly within grid boundaries.
- It checks collision with pacman, which is used for lose condition.
- Our game has two ghost objects

Pacman Class:

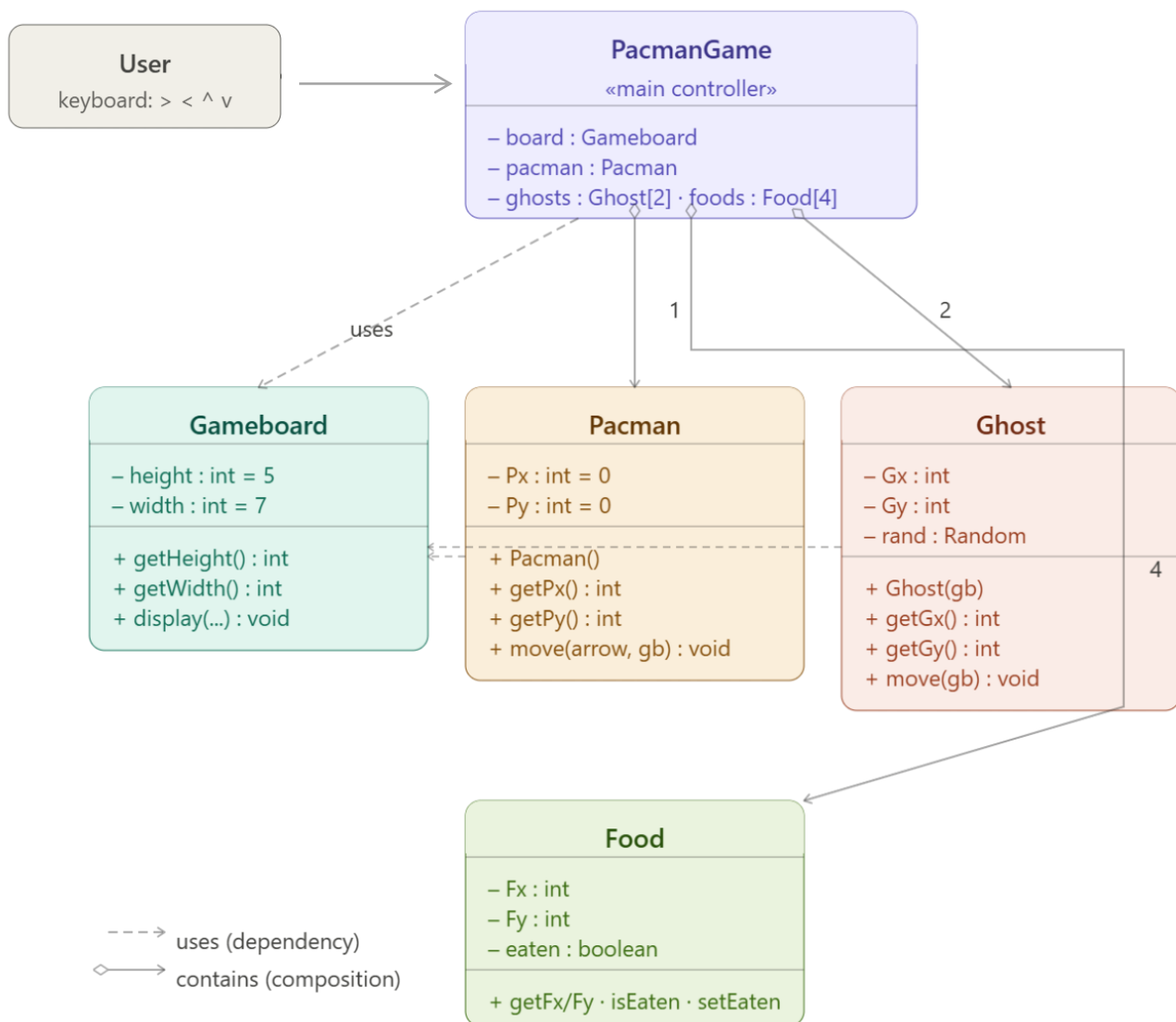
- It stores coordinates of pacman on grid.
- It moves according to string input from user.
- It checks collisions with food and ghost.

Game Logic:

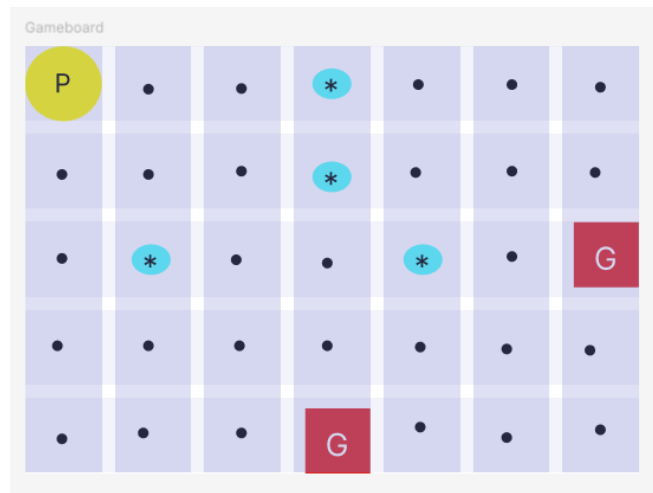
The game loops as long as neither the win nor lose conditions are met. The game board is displayed, user input is taken and all object positions are updated. The pacman is controlled by user input, Ghosts move randomly and food

objects are placed randomly on grid. Collision detection is run every loop to check win or lose conditions.

Game Architecture:



Console view:



Conclusion:

This project demonstrates object-oriented programming concepts such as classes, objects, encapsulation, and interaction between multiple entities. It also implements basic game mechanics such as movement, collision detection, and win/lose conditions in a grid-based environment.